

interpretation, making it better suited for biological sequences and enabling integration with experimental data. For model training, we introduced the local similarity loss, defined as the L1 norm of latent feature differences between neighboring residues, sharing the same form as the L1 norm used for enforcing sparsity. Adding this loss not only preserves sparsity in the latent space and maintains the quality of reconstructed embeddings, but also promotes coherent activation of latent features, thereby facilitating more effective identification of functional motifs. Current local similarity loss operates at the sequence level, incorporating three-dimensional proximity information could enable MotifAE to capture more complex structural motifs. Interpreting latent features is the most challenging part. A common approach is to compare latent feature activation patterns with known functional sites, such as those annotated in UniProt (as in Simon et al.³⁹) or ELM (as used here). However, this strategy is limited by the incompleteness of current annotations and cannot reveal novel motifs, as some latent features may capture functions absent from existing annotations. Although Simon et al.³⁹ employed large language models to further annotate latent features, this approach still relied on known protein functions described in literature. Our work provides two strategies for annotating and potentially discovering novel motifs. First, we calculated PSSMs for each MotifAE feature; features exhibiting well-defined sequence patterns (high KL divergence; **Figure 5C-D**) may correspond to novel motifs. Future studies are needed to systematically link these sequence patterns to biological functions or protein properties. Second, our MotifAE-G framework can be applied to experimental datasets that lack explicit site-level annotations, and the activated residues of selected features may reveal the mechanistic basis of the function or property measured in the experiment.

Looking forward, MotifAE could be further improved in several directions. First, we currently use the L1 norm for the sparsity loss because it shares the same form as the local similarity loss. Alternative sparsity-promoting strategies, such as TopK⁴⁰ and BatchTopK⁴¹ approaches, have been shown to improve SAE training, and future work is needed to explore how these can be integrated with the local similarity constraint. Second, since deep learning models inherently bias to their training data, both the pre-training dataset of the underlying pLM and the dataset used to train MotifAE influence the biological signals captured by latent features. Investigating how dataset composition shapes latent features is important. Finally, applying MotifAE-G to other large-scale functional assays, such as enzymatic activity and protein degradation profiling, could link MotifAE features to other functions.

Overall, we established a framework for the unsupervised discovery and interpretation of functional motifs from pLMs, enabling the study of motifs at an evolutionary scale. By systematically uncovering the sequence determinants of protein function, MotifAE provides a versatile tool for protein annotation, mutational effect interpretation, and rational protein engineering.

Methods

Sparse autoencoder architecture

Sparse autoencoder (SAE) projects language model embeddings into a sparse latent space. The model can be formulated as:

$$f = \text{ReLU}(W_{\text{encoder}}(x - b)) \quad (1)$$

$$\hat{x} = W_{\text{decoder}} \cdot f + b \quad (2)$$

where W_{encoder} projects the original embedding x into the high dimensional latent space f and W_{decoder} does the reverse to get the reconstruction $\hat{x}$. b is the bias term. ReLU was used as the activation function, setting negative values to zero and retaining positive values. Both the encoder and decoder consist of a single linear layer.

Here, the 650-million-parameter version of ESM2 was used, which produces embeddings with a dimensionality of 1,280. The SAE and MotifAE models were trained with varying latent space dimensions, and a dimension of 40,960 was selected (**Figure S1**).

MotifAE loss function

MotifAE employs three losses: the reconstruction loss (L_{rec}), the sparsity loss (L_{sp}), and the local similarity loss (L_{local}). The reconstruction and sparsity losses follow the SAE framework, preserving essential information from the input embeddings while enforcing sparse feature activations. The local similarity loss further constrains at least one nearby residue to have similar latent features, encouraging coherent activation patterns that capture the sequential continuity of protein motifs and basic structural elements. The loss functions are defined as follows:

$$L_{rec} = \frac{1}{n} \sum_{i=1}^n \|x_i - \hat{x}_i\|_2^2 \quad (3)$$

$$L_{sp} = \frac{1}{n} \sum_{i=1}^n |f_i| \quad (4)$$

$$D_i = \{|f_{i+1} - f_i|, \dots, |f_{i+k} - f_i|\} \quad (5)$$

$$L_{local} = \frac{1}{n-k} \sum_{i=1}^{n-k} \text{softmax}(D_i) \cdot D_i \quad (6)$$

$$L_{total} = L_{rec} + \alpha \times L_{sp} + \beta \times L_{local} \quad (7)$$

Here, i denotes the residue index; n denotes the protein length; x and $\hat{x}$ represent the raw and reconstructed embeddings, respectively; and f denotes the latent feature activations. D_i quantifies the L1-norm difference in latent feature activations between residue i and its k neighboring residues. L_{local} encourages at least one nearby residue to have a similar latent feature activation. To achieve this, we apply a softmax over D_i to approximate the minimum difference. Only one direction along the sequence is considered for each residue, but since L_{local} is computed for the full sequence, the opposite direction is accounted for when evaluating neighboring residues. k is the local window size used for similarity calculation, which we set to three.

Model training

MotifAE and SAE were trained and evaluated on 2.3 million representative protein sequences clustered from the AlphaFold Protein Structure Database using Foldseek⁴² (downloaded from <https://afdb-cluster.steineggerlab.workers.dev/>). Of these, 2.2 million sequences were used for training and the remainder for evaluation. Since ESM2 was trained with a maximum sequence length of 1,024, proteins exceeding this length were randomly truncated to a 1,024-residue region.

Models were implemented in PyTorch⁴³ (v2.2) and optimized using the Adam optimizer with a maximum learning rate of 0.001 and a 500-step linear warm-up. A batch size of 40 proteins was used. Residue order was preserved during training to retain positional information required by the local similarity loss. MotifAE was trained with both sparsity and local similarity loss weights set to 0.4, whereas the standard SAE was trained with a sparsity loss weight of 0.85 and no local similarity loss. The weights of the sparsity and local similarity losses were gradually annealed over 5,000 steps. Training was conducted for 80,000 optimization steps on a single NVIDIA L40S GPU, requiring approximately 10 hours per model.

Mutation effect prediction and data processing

The wild-type marginal method²² was used to calculate log-likelihood ratio (LLR) to estimate mutation effects, which was calculated as:

$$LLR_i^{mt} = \log(p(x_i^{mt}|\mathbf{x})) - \log(p(x_i^{wt}|\mathbf{x})) \quad (8)$$

Where i represents residue index, mt and wt represent the mutant and wild-type amino acids, $\mathbf{x}$ is the full sequence without mask.

The ProteinGYM²³ database consists of over 200 protein deep mutational scanning (DMS) experiments, covering mutation effects across diverse protein functions. In this study, only single substitution mutations were evaluated.

The ClinVar²⁶ database provides expert-annotated mutations related to human diseases. Data preprocessing followed the procedure described in our previous work⁴⁴. For this study, we considered only genes containing at least five pathogenic and five benign mutations, equal number of pathogenic and benign mutations were randomly sampled from each protein, resulting in a gene-balanced dataset of 2,272 pathogenic and 2,272 benign mutations in 207 genes. We used this dataset because ClinVar exhibits strong gene-level bias: the ratio of pathogenic to benign mutations varies substantially among genes⁴⁵.

For protein stability³², the dataset “Tsuboyama2023_Dataset2_Dataset3_20230416.csv” and AlphaFold2-predicted protein structures were downloaded from <https://zenodo.org/records/7992926>. Proteins were defined using the “WT_name” column in the table. Only single substitution mutations with “ddG_ML” values were analyzed. The

“ddG_{ML}” values for the same sequence were averaged. Wild-type sequences of 412 proteins were clustered using MMseqs2⁴⁶ easy-cluster with the parameters: `--min-seq-id 0.3 -c 0.5 --cov-mode 1`. Homologs in the UniRef50 (downloaded in February 2025) were identified using MMseqs2 search with the parameters: `-s 7 -a 1`. Solvent accessibility was computed from predicted structures using mdtraj⁴⁷. The raw solvent-accessible surface areas were normalized by the maximum accessible area of each amino acid residue, resulting in normalized values ranging from 0 to 1. For MotifAE-G training, 70% mutations in training proteins were used to select gates, while the remaining 30% were used as validation set to optimize hyperparameters.

Evaluation on ELM motifs and homodimer

ELM motifs were downloaded from the ELM² database in January 2025, and only instances annotated as *true positive* were analyzed. For proteins longer than 1,024 residues, a subsequence of length 1,024 was selected with the motif region positioned at the center. IUPred3²⁸ was used to predict disordered regions: residues with both long disordered region prediction > 0.5 and short disordered region prediction > 0.5 were classified as disordered, whereas all others were classified as ordered.

PINDER³⁰ is a database of PDB protein complexes. We filtered the dataset to include only structures released after September 30, 2021, and conducted data quality control following the PINDER supplementary material. From the resulting dataset, we got 3,303 homodimers and annotated contact residues as those with heavy atoms within 5 Å. We further removed redundancy by clustering the filtered dataset at a sequence identity threshold of 30% using MMseqs2 and retaining only the representative sequence from each cluster. Homodimers with less than five contacts were excluded. The final dataset comprised 1,565 dimers, with a median number of contacts of 44.

These two tasks were formulated as a binary classification problem, where each residue is labeled as either part of a motif/interface or not. In MotifAE and SAE, each residue is represented by a 40,960-dimensional feature vector. For each latent feature, its activation value was used as the residue-level prediction of motif/interface to calculate AUROC.

Activated peptides analysis and motif calculation

Activated peptides for each MotifAE feature were identified across the 2.3 million representative proteins. For each protein, only features with a summed activation value > 1 were analyzed. Coherent activations were recorded, and activations separated by a gap of one or two amino acids were merged. Only activated peptides with a mean feature activation value > 0.2 and lengths between 5 and 30 residues were retained for analysis.

To calculate position-specific scoring matrices (PSSMs), up to 1,000 activated peptides ranked by mean activation value were used for each feature. GibbsCluster 2.0³¹ was employed to align the activated peptides and remove outliers with parameters: `-g 1 -l 5 -T -j 1 -l 1 -D 1`. The motif length was set to 5 amino acids. A trash cluster was used to remove sequences that did not align well with others, with the threshold set to 1. The alignment cores of length 5 generated by GibbsCluster were subsequently used to construct PSSMs, which were visualized using the Logomaker⁴⁸ Python package.

MotifAE-G architecture and training

The MotifAE-G model introduced a gating mechanism to amplify or attenuate feature signals during embedding reconstruction, which were subsequently used for downstream tasks. The gate was applied to each latent feature of MotifAE as:

$$f' = f \odot gate \quad (9)$$

where $\odot$ denotes elementwise multiplication.

For protein stability prediction, a learnable binary gate was introduced while keeping ESM2 and MotifAE parameters fixed. Each gate value could only take 0 or 1, and only features with a gate value of 1 were retained for reconstructing the embeddings. The gate was binarized using the Straight-Through Estimator^{49,50}. In the forward pass, each gate was discretized as:

$$gate = \begin{cases} 1 & \sigma(g) > 0.5 \\ 0 & \sigma(g) \leq 0.5 \end{cases} \quad (10)$$

while in the backward pass, gradients were propagated directly through the sigmoid output $\sigma(g)$.